Learning Dynamic Protein Representations at Scale with Distograms

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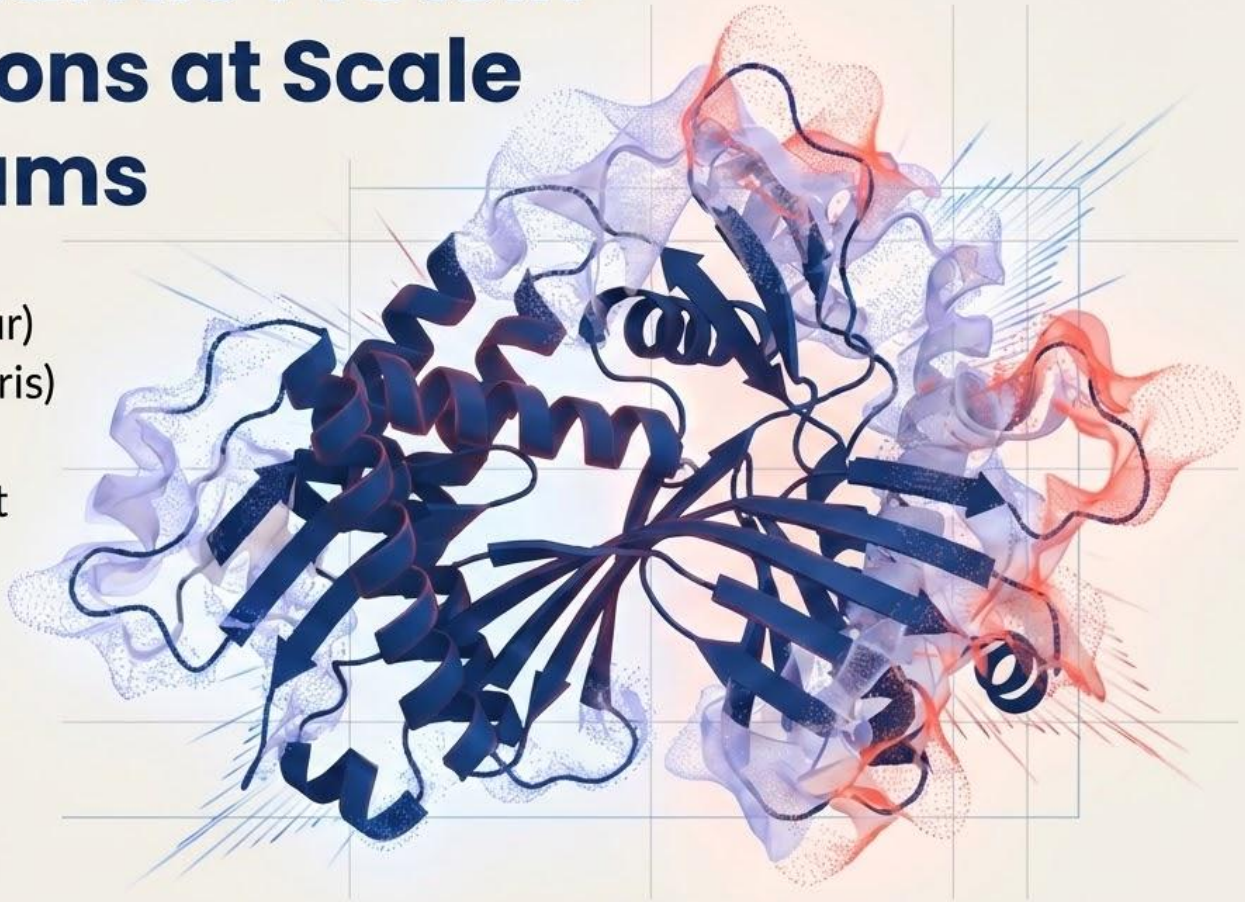

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LifeLU Reading Club

Why Dynamics Matter

Protein function depends on **conformational flexibility**:

- ligand binding
- allosteric regulation
- structural rearrangements

However, most ML models use **single static structures**.:

Representation learning often **ignores structural variability**.

Current Solutions and Their Limits

Physics-based solution:

- Molecular Dynamics (MD)
- Sample from Boltzmann distribution
- Accurate but:
  - Extremely expensive
  - Not scalable

Generative models:

- Diffusion / flow / ensemble generators
- Still:
  - Data-hungry
  - Incomplete exploration

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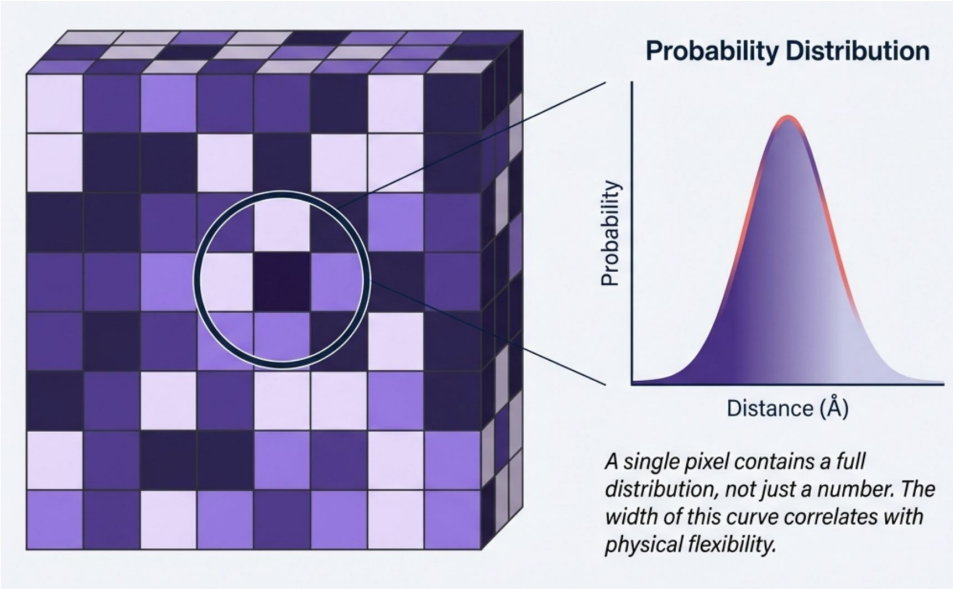
Can we encode dynamics ***without generating trajectories?***

Core Idea: *Distograms instead of MD*

Distogram

- Predicted residue–residue distance probability distribution
- Outputs
  - AlphaFold
  - Boltz2



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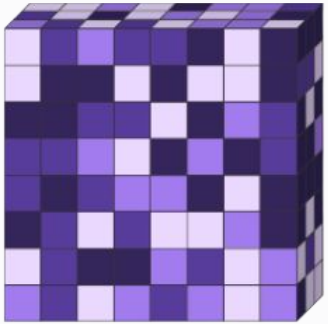


Distograms encode:

- Expected distance
- Structural **uncertainty**
- Potential conformational **variability**

Cheap to compute
→ Orders of magnitude ***faster than MD***

Overview

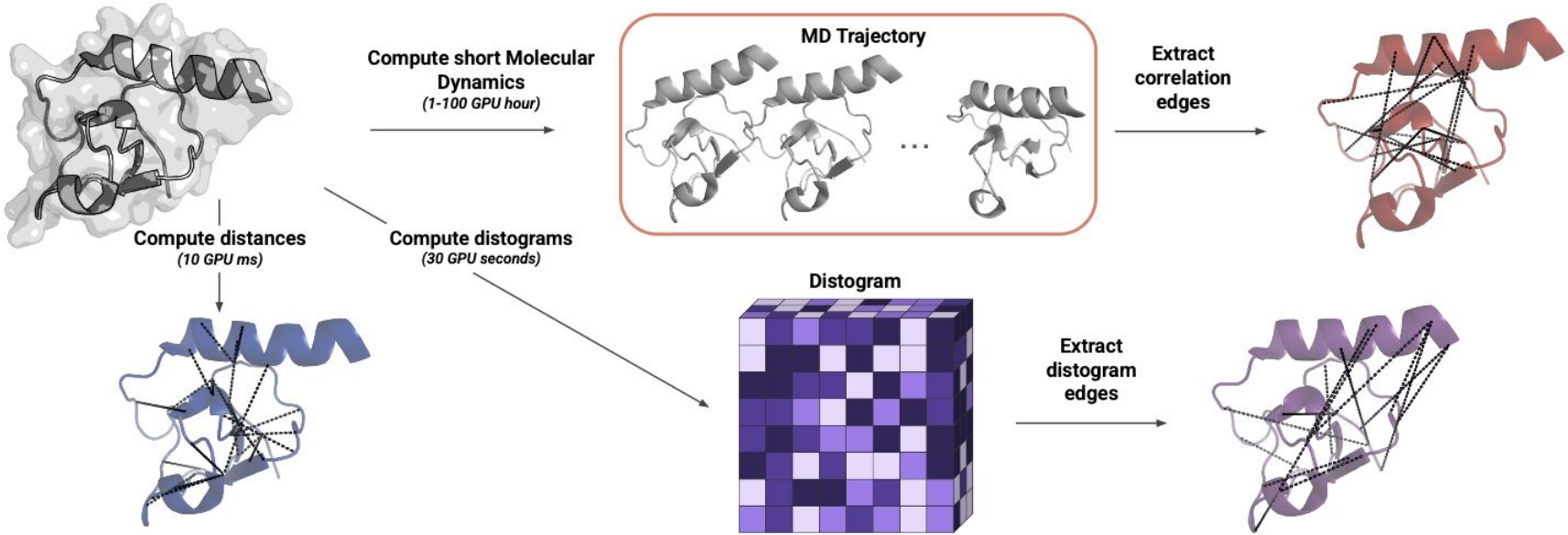


Figure 1. Overview of the protein representations used in this study. A protein can be represented as a static structure, computed on the fly. To capture dynamics, previous work often relies on compute-intensive MD simulations, encoding the results as additional edges. In contrast, we propose bypassing simulations by leveraging predicted distograms to incorporate dynamic information.

Graph Construction

Base graph:

Nodes:

- Atoms
- Residues (Cα)

Node features:

- one-hot encoding of the amino acid type or atom type

Edges:

Radius cutoff: Distance

$$\mathcal{E}_{dist} = \{(v_i, v_j) \mid d(v_i, v_j) < \tau_{dist}\},$$

MD: Residue motion correlation

$$\mathcal{E}_{corr} = \{(v_i, v_j) \mid |C_{ij}| > \tau_{corr}\},$$

Random: for long-range MP

$$\mathcal{E}_{rand} \sim \mathcal{U}(\mathcal{V} \times \mathcal{V}),$$

Distogram-based edge features

- Distribution of the distance between each pair of residues (vi, vj),
- Represent prediction uncertainty and, possibly, variability due to the underlying dynamics
- Discretized into B bins

Distogram edges

The value at position (i, j, k) corresponds to the predicted probability that the distance between nodes v_i and v_j falls into bin k, i.e. $\mathcal{D}(i, j, k) = \mathbb{P}[b_k \leq d(v_i, v_j) < b_{k+1}]$.

$$\mathcal{E}_{\text{disto}} = \{(v_i, v_j) \mid \mathbb{P}[d(v_i, v_j) \leq \delta] > \tau_{\text{disto}}\},$$

τ_{disto} probability threshold
δ distance cutoff for the distogram neighbors

Pair specific distance cutoff

$$\delta_{u,v} = \mu_{u,v} + 1.645\sigma_{u,v},$$

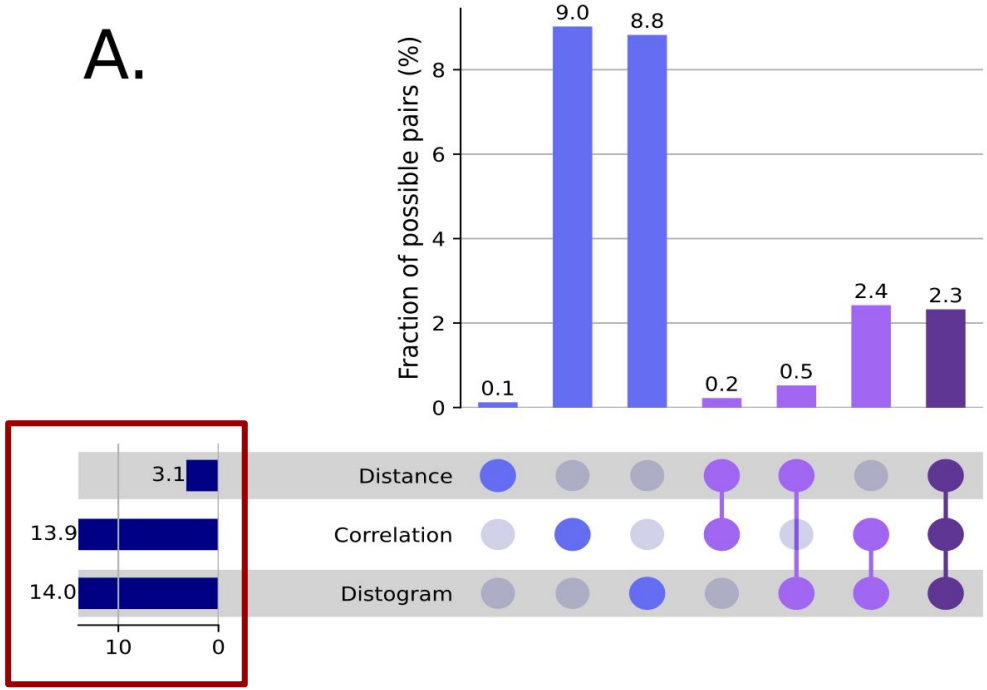
translated into the closest cutoff bin

$$\mathbb{P}[d(v_i, v_j) \leq \delta_{v_i,v_j}] = \sum_{k \leq b_\delta^{v_i,v_j}} \mathcal{D}(i, j, k).$$

Edge definitions capture related but distinct signals

Edge coverage

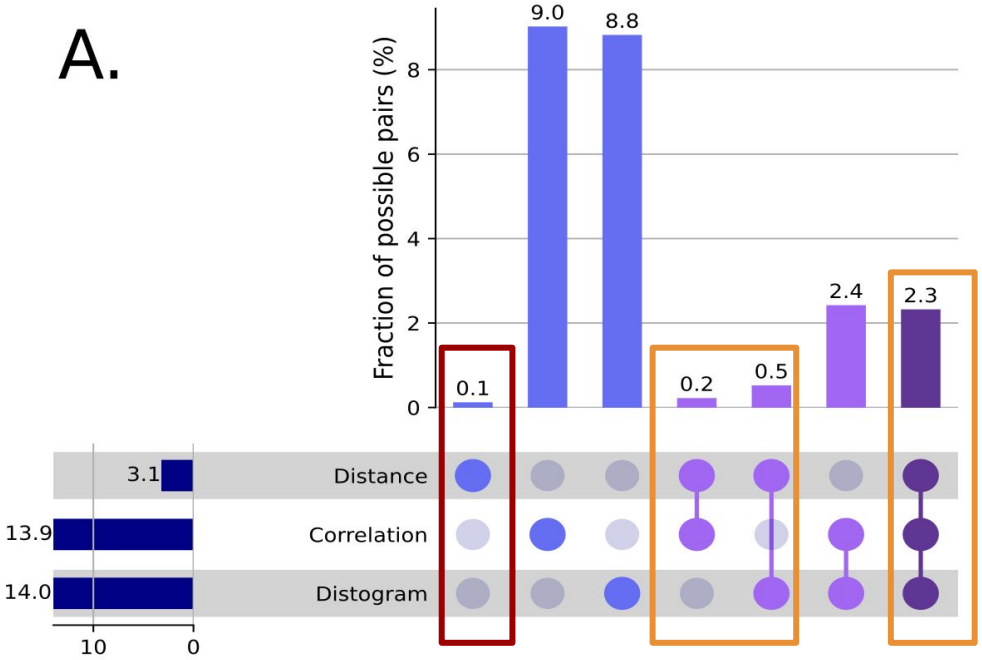
- Distance edges: **3.1%** of residue pairs
- Correlation edges: **13.9%**
- Distogram edges: **14.0%**



Edge definitions capture related but distinct signals

Strong coverage of structural contacts

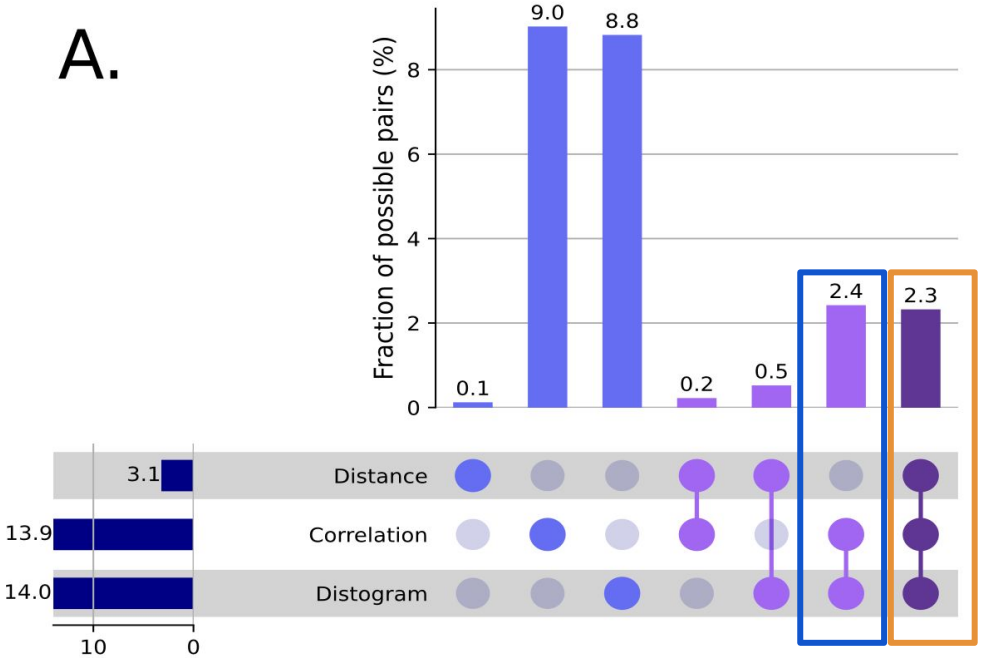
- Only **0.1%** of residue pairs are **distance-specific**
- Most spatial contacts are **also captured by distogram or correlation edges**



Edge definitions capture related but distinct signals

Moderate agreement between correlation and distogram

- **2.4%** overlap between correlation and distogram edges
- **2.3%** shared by all three edge types



Edge definitions are **not redundant;**
they capture **complementary structural relationships.**

Correlation and Distogram Edges Enable Long-Range Communication

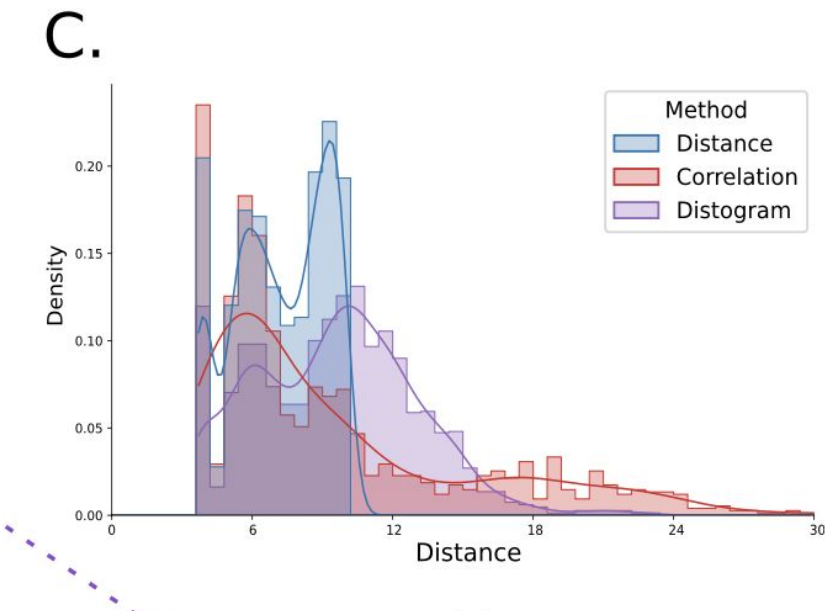
Dataset-wide trend

Across the entire dataset:

- **Correlation edges:** 17.1 Å
- **Distogram edges:** 19.2 Å
- **Distance edges:** 7.2 Å

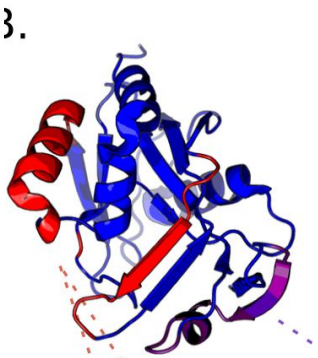
Case Study: Protein 6NVD

- **Distance edges (Edist):** 7.16 Å
  Correlation edges (Ecorr): 9.63 Å
- **Distogram edges (Edisto):** 9.57 Å



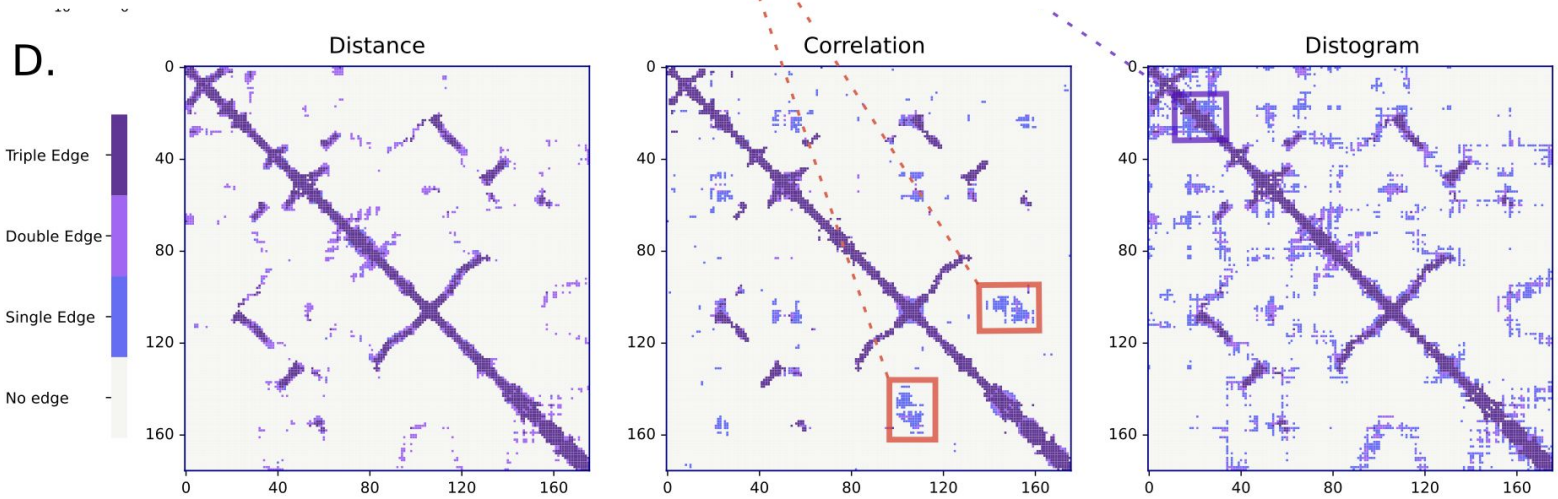
Edges specific to **correlation graphs** map to residues located in:

- a **helix**
- a **β-sheet motif**



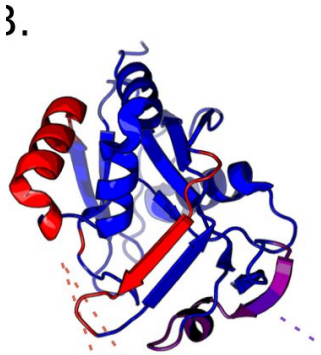
These elements **move in a coordinated way** during dynamics.

- **not necessarily spatially close**
- reflect **coupled motion rather than proximity**



Distogram edges capture structural uncertainty

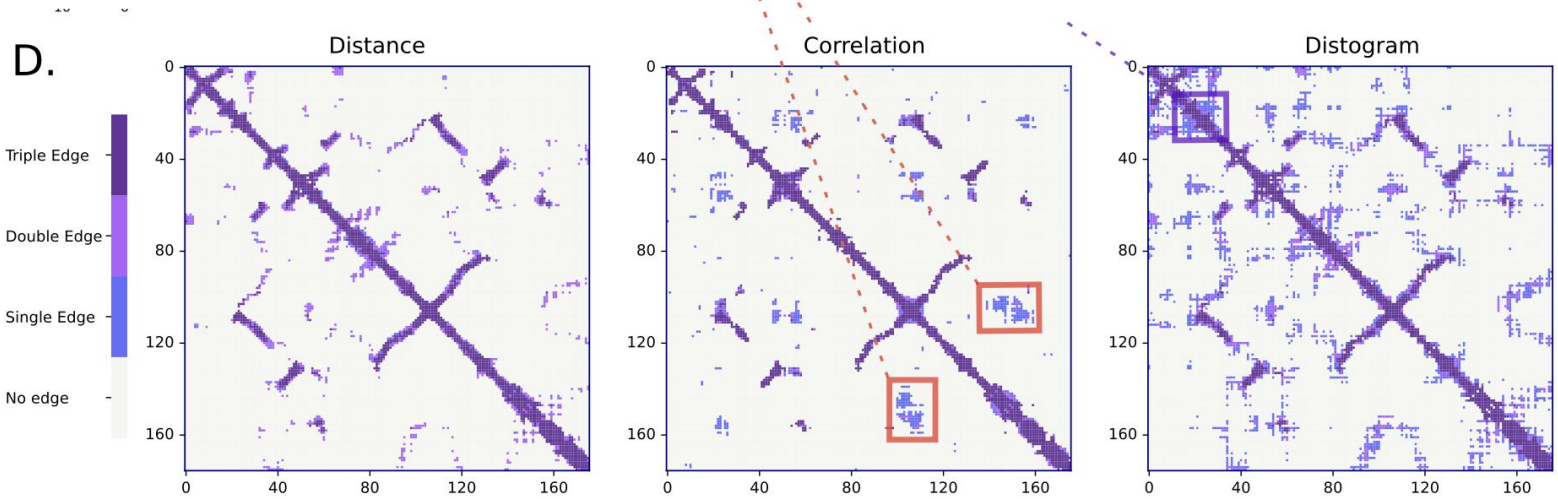
Distogram edges tend to appear **around distance edges**, forming local clusters.



Regions correspond to:

- **Flexible loops**
- **Dynamically changing secondary structures**

These edges likely encode **distance fuzziness or structural variability**



Why Relational GNN?

Multiple edge types:

- Distance
- Correlation
- Distogram
- Random (control)

Architectures:

- R-GCN
- R-GAT
- R-EGNN

Each edge type = separate relation channel

Use R-GNN:

$$h_i^{(l+1)} = \sigma\left(W_0^{(l)} h_i^{(l)} + \sum_{r\in\mathcal{R}} \sum_{j\in\mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)}\right)$$

Relational GNNs

R-GCN

Relation-specific weight matrices

$$m_{r,i}^{l} = W_r^{(l)} h_i^{(l)}.$$

$$h_i^{(l+1)} = \sigma\left(m_{0,i}^{(l)} + \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} m_{r,j}^{(l)}\right),$$

R-GAT

Relation-specific attention coefficients to weight messages from different neighbors

$$h_i^{(l+1)} = \sigma\left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \alpha_{ij}^r m_{r,j}^{(l)}\right)$$

$$\alpha_{ij}^r = \frac{\exp\left(\text{LeakyReLU}\left(a_r^\top [m_{r,i}^{(l)} \,||\, m_{r,j}^{(l)}]\right)\right)}{\sum_{k \in \mathcal{N}_i^r} \exp\left(\text{LeakyReLU}\left(a_r^\top [m_{r,i}^{(l)} \,||\, m_{r,k}^{(l)}]\right)\right)};$$

Relational GNNs

R-EGNN

a relational variant of EGNN by processing each relation type separately

$$m_{ij}^r = \phi_e(h_i^{(l)}, h_j^{(l)}, \|x_i^{(l)} - x_j^{(l)}\|^2, e_{ij}^r),$$
$$\Delta x_i^r = \frac{1}{|\mathcal{N}_i^r|} \sum_{j \in \mathcal{N}_i^r} (x_i^{(l)} - x_j^{(l)}) \phi_x(m_{ij}^r),$$
$$h_i^{(l+1),r} = \phi_h\Big(h_i^{(l)}, \sum_{j \in \mathcal{N}_i^r} m_{ij}^r\Big),$$

node representation and coordinates update:

$$h_i^{(l+1)} = \sum_{r \in \mathcal{R}} h_i^{(l+1),r},$$
$$x_i^{(l+1)} = x_i^{(l)} + \sum_{r \in \mathcal{R}} \Delta x_i^r.$$

Experimental Setup

Dataset

- MISATO: short MD trajectories

Two tasks

- Ligand binding site prediction (443 nodes avg)
- Binding affinity regression (47 nodes avg)

Compare:

- Distance only
- Correlation (MD-based)
- Random
- Distogram
- Distogram features

Distogram-based edges and features improves performance across tasks and architectures

Binding Site Prediction (Mean number of nodes = 443)					
Model	Graph Type	Accuracy (↑)	Precision (↑)	Recall (↑)	F1 score (↑)
R-GCN	Distance	0.761	0.187	0.420	0.259
	Distance + Correlation	**0.809**	0.217	0.357	0.270
	Distance + Random Edges	0.773	0.193	0.405	0.261
	Distance + Distogram	0.782	**0.236**	**0.535**	**0.328**
R-GAT	Distance	0.753	0.196	0.478	0.278
	Distance + Correlation	0.791	0.229	0.469	0.308
	Distance + Random Edges	0.760	0.217	0.546	0.311
	Distance + Distogram	0.814	0.289	0.598	0.390
	Distance + Distogram + Features	**0.816**	**0.298**	**0.627**	**0.404**
R-EGNN	Distance	0.832	0.282	0.444	0.345
	Distance + Correlation	**0.882**	**0.393**	0.350	0.370
	Distance + Random Edges	0.830	0.316	**0.607**	0.416
	Distance + Distogram	0.861	0.376	0.606	**0.464**
	Distance + Distogram + Features	0.859	0.371	0.602	0.459

Distogram > Correlation-based MD edges > Random

Distogram-based edges and features improves performance across tasks and architectures

Model	Graph Type	MAE (↓)	RMSE (↓)	Pearson R (↑)	Spearman R (↑)
		Binding Affinity Prediction (Mean number of nodes = 47)			
R-GCN	Distance	1.244	1.562	0.689	0.656
	Distance + Correlation	1.299	1.601	0.673	0.637
	Distance + Random Edges	**1.171**	**1.453**	**0.744**	**0.718**
	Distance + Distogram	1.195	1.487	0.731	0.694
R-GAT	Distance	1.226	1.562	0.691	0.658
	Distance + Correlation	1.280	1.568	0.686	0.656
	Distance + Random Edges	1.236	1.542	0.701	0.685
	Distance + Distogram	**1.176**	**1.487**	**0.740**	**0.713**
	Distance + Distogram + Features	1.179	1.492	0.722	0.693
R-EGNN	Distance	1.296	1.623	0.666	0.642
	Distance + Correlation	1.357	1.713	0.611	0.576
	Distance + Random Edges	1.211	1.502	0.721	0.698
	Distance + Distogram	1.275	1.560	0.699	0.674
	Distance + Distogram + Features	**1.208**	**1.479**	**0.736**	**0.725**

Random ~ Distogram

Small graphs > well-connected

Distogram-based edges and features improves performance across tasks and architectures

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MD-correlation does NOT consistently improve

Comparison to MD-correlation based approaches

Observations

- **Not able to reproduce the gains** from correlation edges reported by Guo et al. (2025).

- **Distance-only models perform strongly**, and adding correlation edges **does not consistently help**.

- In some cases, **random edges outperform correlation edges**.

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Key Insight

- **Distograms capture structural information beyond:**
  - static distance thresholds
  - correlation-based neighborhoods

Result

- Improved performance for:
  - **Binding site prediction** (residue level)
  - **Binding affinity prediction** (atom level)

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Key Insight

- **Distograms capture structural information beyond:**
  - static distance thresholds
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Distogram edges provide **richer structural context** than **distance or correlation edges** while avoiding expensive MD simulations.

or:
on
(residue level)
- **Binding affinity prediction** (atom level)

Distograms encode stability-relevant variability, complements the pretrained ProteinMPNN embeddings on stability prediction

Baseline: ThermoMPNN - ProteinMPNN

Add:

- Distogram edge features (small proteins (avg 65 nodes), 48 nearest neighbor in ProteinMPNN))
- Adapter layer & finetuning

Connectivity	Distogram Features	R^2 (↑)	RMSE (↓)	Spearman R (↑)	Pearson R (↑)
Baseline (Topk = 48)		0.518	0.727	0.726	0.761
Baseline	✓	**0.577**	**0.681**	**0.749**	**0.788**
Ablation (Topk = 16)	✓	0.550	0.702	0.727	0.770
Ablation + 8 Random Edges	✓	0.556	0.698	0.733	0.773
Ablation + 8 Distogram Edges	✓	**0.557**	**0.697**	**0.740**	**0.778**

Adding distogram edge features results in improvements, across different baseline connectivity and tasks.

RNA molecules typically display **greater flexibility** due to less hydrophobic nature

- RNA-CM (chemical modification)
- RNA-Site (binding site)

Adding distogram features: **consistent improvements** across graph types

Interesting: RNA structure prediction is **less accurate**
→ Yet distograms still useful

Graph Type	Distogram Features	Balanced Accuracy (↑)
RNA-CM (Mean number of nodes = 47)		
Distance		52.8
	✓	**54.8**
2D+		64.5
	✓	**65.5**
2.5D		66.7
	✓	**67.3**
RNA-Site (Mean number of nodes = 64)		
Distance		59.1
	✓	**59.3**
2D+		61.2
	✓	**62.1**
2.5D		**61.9**
	✓	60.8

Table 3. Performance of distogram-based edge features on RNA property prediction tasks. We introduce these edges in three different graph constructions: Distance-based, 2D+ and 2.5D.

What Makes This Powerful?

Advantages:

- No MD needed
- Scalable
- Generalizable
- Easy to integrate into existing GNNs
- Works for proteins and RNA

Distograms are:

- Already available
  Cheap
- Under-exploited

Discussion

- Relies on high-quality distograms
  → MSA quality dependent
- Large proteins expensive
- Only pairwise marginalized distributions
  → No higher-order correlations
- Apply to proteins without experimental structures
- Test on low-confidence AlphaFold predictions

Big Picture & Takeaways

Protein function depends on dynamics, but MD simulations are **expensive and hard to scale**.

Key idea: We don't need explicit simulations to capture dynamics.
Distograms encode dynamic information implicitly.

Distograms act as **uncertainty representations / implicit ensembles** of residue distances.

Result: Distogram-based graphs **outperform MD-correlation graphs**.

Distograms provide a **scalable bridge** between static structures and dynamic-aware protein modeling.